Answer key

SAT Practice Test 5, Reading and Writing

Module 1

NO.	ANSWER	SKILL
1	**D**	Words in context
2	**A**	Words in context
3	**B**	Words in context
4	**C**	Words in context
5	**C**	Words in context
6	**B**	Text structure and purpose
7	**A**	Text structure and purpose
8	**D**	Text structure and purpose
9	**A**	Central ideas and details
10	**C**	Central ideas and details
11	**D**	Cross-text connections
12	**B**	Command of evidence: quantitative
13	**B**	Command of evidence: quantitative
14	**C**	Command of evidence: textual
15	**D**	Command of evidence: textual
16	**A**	Command of evidence: textual
17	**C**	Inferences
18	**B**	Inferences
19	**D**	Inferences
20	**A**	Boundaries
21	**C**	Form, structure, and sense
22	**D**	Form, structure, and sense
23	**A**	Form, structure, and sense
24	**B**	Boundaries
25	**C**	Form, structure, and sense
26	**B**	Form, structure, and sense
27	**D**	Boundaries
28	**A**	Transitions
29	**A**	Transitions
30	**B**	Transitions
31	**C**	Rhetorical synthesis
32	**D**	Rhetorical synthesis
33	**A**	Rhetorical synthesis

Module 2

NO.	ANSWER	SKILL
1	**D**	Words in context
2	**A**	Words in context
3	**C**	Words in context
4	**B**	Words in context
5	**C**	Words in context
6	**A**	Text structure and purpose
7	**D**	Text structure and purpose
8	**B**	Text structure and purpose
9	**D**	Cross-text connections
10	**C**	Central ideas and details
11	**A**	Central ideas and details
12	**B**	Command of evidence: quantitative
13	**B**	Command of evidence: quantitative
14	**A**	Command of evidence: textual
15	**C**	Command of evidence: textual
16	**D**	Command of evidence: textual
17	**C**	Inferences
18	**D**	Inferences
19	**B**	Inferences
20	**A**	Boundaries
21	**C**	Form, structure, and sense
22	**A**	Boundaries
23	**D**	Form, structure, and sense
24	**B**	Form, structure, and sense
25	**A**	Boundaries
26	**D**	Form, structure, and sense
27	**C**	Boundaries
28	**B**	Transitions
29	**A**	Transitions
30	**D**	Transitions
31	**B**	Rhetorical synthesis
32	**C**	Rhetorical synthesis
33	**D**	Rhetorical synthesis

Answer key

SAT Practice Test 5, Math

Module 1

NO.	ANSWER	SKILL
1	**C**	Data analysis
2	**A**	Linear equations
3	**D**	Linear equations in context
4	**B**	Systems of equations
5	**B**	Functions
6	**C**	Nonlinear expressions
7	**A**	Ratios and rates
8	**D**	Triangles
9	**C**	Circles
10	**B**	Quadratics
11	**D**	Probability
12	**A**	Statistics
13	**C**	Scatterplots
14	**D**	Inequalities
15	**A**	Functions
16	**B**	Exponential functions
17	**C**	Linear functions
18	**D**	Percent change
19	**A**	Volume
20	**B**	Right triangles
21	**10**	Quadratics
22	**2**	Linear equations
23	**48**	Linear equations
24	**25**	Systems of equations
25	**15**	Right triangles
26	**15**	Percent change
27	**396**	Volume

Module 2

NO.	ANSWER	SKILL
1	**D**	Percentages
2	**A**	Volume
3	**C**	Inequalities
4	**B**	Statistics
5	**A**	Functions
6	**B**	Probability
7	**D**	Systems of equations
8	**C**	Equivalent expressions
9	**D**	Ratios and rates
10	**C**	Absolute value
11	**A**	Percentages
12	**B**	Statistics
13	**C**	Systems of equations
14	**A**	Linear functions in context
15	**B**	Surface area
16	**D**	Functions
17	**D**	Exponential equations
18	**A**	Percentages
19	**C**	Exponential functions
20	**B**	Equivalent expressions
21	**28**	Systems of equations
22	**36**	Quadratics
23	**4**	Systems of equations
24	**60**	Similar triangles
25	**7**	Circles in the plane
26	**6**	Quadratics
27	**5**	Systems with nonlinear equations

Score yourself. Count the questions you got right in each module.

Reading and Writing, Module 1: ___ of 33

Module 2: ___ of 33

Math, Module 1: ___ of 27

Module 2: ___ of 27

Answer explanations

SAT Practice Test 5, Reading and Writing, Module 1

Questions 1 to 17

1 **Choice D** **Words in context**

A trail far steeper than expected would dismay hikers. Delighted and relieved suggest pleasure, and indifferent does not fit the contrast with the map.

2 **Choice A** **Words in context**

The colon explains that stone survives thousands of years, which is durability. Fragile is the opposite, and the passage does not mention price or portability.

3 **Choice B** **Words in context**

Her cheerful manner contrasted with the grim news, so it belied it. Underscored and reflected would match the seriousness.

4 **Choice C** **Words in context**

Though sets up a contrast with the later criticism, so the first volume was received with acclaim. Scorn would not contrast.

5 **Choice C** **Words in context**

A calm tone in the face of raised voices shows composure. Agitated is the opposite.

6 **Choice B** **Text structure and purpose**

The underlined sentence gives examples of different flash patterns, supporting the claim that patterns are specific to each species. It does not dispute the claim, or discuss females not flashing or food.

7 **Choice A** **Text structure and purpose**

The speech weighs on Isidore like a stone, but at the end he realises he does not need it, which shows a shift from anxiety to confidence. The passage does not criticise the guests, and he is not unprepared.

8 **Choice D** **Text structure and purpose**

The first part describes the rain, the woman and the child, and the last lines reflect that waiting is something people do and let pass. No complaint, destination or childhood memory appears.

9 **Choice A** **Central ideas and details**

The text reports that much of the benefit seemed to be relief of withdrawal. It does not say decaffeinated coffee is better, that drinkers always know their dose, or that caffeine has no effect on anyone.

10 **Choice C** **Central ideas and details**

The text says English borrowed words with foods but that words also travel in the other direction. It makes no claim of most borrowings, about how often foods are named, or that borrowed words are never changed.

11 **Choice D** **Cross-text connections**

Text 2 says height limits raise prices by limiting supply and proposes design rules instead. The economist does not ignore character or deny that low-rise streets encourage walking.

12 **Choice B** **Command of evidence: quantitative**

Fewer than eight hours includes the 5, 6 and 7 hour groups: 30 + 75 + 105 = 210 students, or 70 percent of 300. The other choices describe the opposite group or a single bar.

13 **Choice B** **Command of evidence: quantitative**

A claim about every district needs data covering all four, and the first choice states that complaints fell in all while the table shows this. The others cover only some districts or contradict the claim.

14 **Choice C** **Command of evidence: textual**

The claim links reduced food to exploration, which the quotation about a few ants leaving the trail shows. The others describe the study, the following behaviour or her interpretation.

15 **Choice D** **Command of evidence: textual**

Longer endurance with music is the direct support. The others describe the design.

16 **Choice A** **Command of evidence: textual**

Removing the rivulets with the coating stopped the vibrations, which indicates they were the cause. The other findings do not isolate rivulets.

17 **Choice C** **Inferences**

Fenced plots lost species, so the elephants' disturbance may maintain diversity. The other choices contradict the findings.

Answer explanations

SAT Practice Test 5, Reading and Writing, Module 1

Questions 18 to 33

18 **Choice B** Inferences

Customers came for the owner's recommendations, and the shop closed when the owner died, so success depended on the owner's knowledge and relationships. The other options are not supported.

19 **Choice D** Inferences

Children learn from peers and the young already merge the vowels, so the merger is likely to spread. The other choices contradict the pattern.

20 **Choice A** Boundaries

The relative clause which lasted nearly an hour is closed by a comma before the main verb left.

21 **Choice C** Form, structure, and sense

Data is conventionally treated as plural in formal writing, so show agrees.

22 **Choice D** Form, structure, and sense

The items in the list must be parallel nouns, so violin, piano, and singing is correct.

23 **Choice A** Form, structure, and sense

The possessive their is needed. There and they're are other words, and its does not agree with the plural hikers.

24 **Choice B** Boundaries

The pronoun is the subject of oversaw, so who is the right case, and the clause is complete with the verb oversaw.

25 **Choice C** Form, structure, and sense

That introduces the noun clause that is the object of announced.

26 **Choice B** Form, structure, and sense

Last Saturday requires the simple past, brought. Brung and bringed are not standard forms, and have brought does not fit the time reference.

27 **Choice D** Boundaries

The second clause contrasts with thoroughness, so nevertheless is correct.

28 **Choice A** Transitions

The slow service contrasts with the excellent food, so However is correct.

29 **Choice A** Transitions

Over time describes what the islanders learned across generations of fishing, so it is the logical transition.

30 **Choice B** Transitions

The reports contradict the promise, so Yet is the right transition.

31 **Choice C** Rhetorical synthesis

To explain the event to a beginner, the choice should give where it began, how long it burned and its toll, which the first does.

32 **Choice D** Rhetorical synthesis

A contrast needs the properties of both materials in terms of the same measure, which the first choice gives.

33 **Choice A** Rhetorical synthesis

Scale is shown by comparing the small group with the huge number of saplings, which the first choice does.

Answer explanations

SAT Practice Test 5, Reading and Writing, Module 2

Questions 1 to 17

1 **Choice D** **Words in context**

Fear of censure suggests the conclusions clashed with the orthodoxy, so at odds fits. In step would give no reason for secrecy.

2 **Choice A** **Words in context**

The surprise at the author's frankness about failures implies that critics expected a boastful, self-congratulatory memoir. Unsparing would match the candour, so it is the opposite of what critics expected.

3 **Choice C** **Words in context**

A gift that transformed the collection is a turning point, which watershed means. Setback and digression imply no improvement.

4 **Choice B** **Words in context**

Contradictory statements make an account inconsistent. Lucid, corroborated and unassailable suggest strength or support.

5 **Choice C** **Words in context**

Keeping every marking and adding none is a scrupulous approach. Cavalier and capricious would imply disregard for the score.

6 **Choice A** **Text structure and purpose**

The underlined sentence shows the problem was serious enough for a large prize, and the next sentence says the solution came from a carpenter. It does not explain latitude, blame the government or describe training.

7 **Choice D** **Text structure and purpose**

Selwyn lets the boy fail in private and nods toward the shore as if nothing had gone wrong, which shows tact and patience. He is not mocking the boy, and the passage does not suggest reluctance to learn or discuss his fishing.

8 **Choice B** **Text structure and purpose**

The poem reports the grandmother's word for the sea and the speaker's younger, narrower view, then reflects that words come from learned work. It contains no prediction, questions or argument.

9 **Choice D** **Cross-text connections**

Text 2 gives a different explanation, pattern detection with plentiful exposure, for the fast learning that Text 1 attributes to inborn grammar. The researcher does not deny fast learning.

10 **Choice C** **Central ideas and details**

The text reports ongoing neuron formation and a possible role in keeping memories distinct. The rate declines, the role is still debated, and blocking the neurons did affect memory.

11 **Choice A** **Central ideas and details**

The text says X-rays show that the artist first painted a tense figure and then revised it to create calm. It does not say the work is a forgery, that he kept the first idea, or that its colours are admired.

12 **Choice B** **Command of evidence: quantitative**

The claim is that the gap was largest at the first attempt, which the first choice shows with the gap and its comparison. The others cover one later attempt, a general statement or a single figure.

13 **Choice B** **Command of evidence: quantitative**

The claim is about both smaller reserves, which the first choice covers with correct figures. The others discuss the larger reserves, one reserve, or the total.

14 **Choice A** **Command of evidence: textual**

The claim ties the clean manuscripts to his method, and only the quotation describing slow composition and setting the sentence down when finished does so. The others state the result or who reported it.

15 **Choice C** **Command of evidence: textual**

Only the combined exposure produced an effect, while each chemical alone had none. This is stated in full only in the first choice. The others give half of the pattern.

16 **Choice D** **Command of evidence: textual**

The economist's evidence is that most new residents worked at the mill. The doubling is the fact to be explained, and historians' views and the railway's arrival are not her evidence.

Answer explanations

SAT Practice Test 5, Reading and Writing, Module 2

Questions 18 to 33

18 Choice D Inferences

Simultaneous independent invention suggests shared conditions made the idea available to many people. The other choices are not supported.

19 Choice B Inferences

The cycle gives true brightness, and comparing it with apparent brightness shows the distance. The text gives no basis for age, number of stars or composition.

20 Choice A Boundaries

The subject of the clause is the plural novels, which, so the plural have been is correct.

21 Choice C Form, structure, and sense

The subject director is singular, and the past tense attended agrees and fits the context.

22 Choice A Boundaries

After Not until, the auxiliary comes before the subject, and the singular cause needs was.

23 Choice D Form, structure, and sense

The list should be parallel noun phrases: careful planning, reliable equipment, and the cooperation of local guides.

24 Choice B Form, structure, and sense

The pronoun is the subject of had been estranged, referring to people, so who is correct.

25 Choice A Boundaries

A colon introduces the explanation of how the members were divided, after a complete clause. A comma would create a comma splice, and the other choices would leave the sentence ungrammatical.

26 Choice D Form, structure, and sense

With two items, the comparative the more detailed is used. The most is superlative.

27 Choice C Boundaries

The twelve volunteers are plural and possess the contributions, so the plural possessive volunteers' is correct. Volunteer's is singular, and the other forms are not possessive or are malformed.

28 Choice B Transitions

The sentence corrects the assumption by supplying the fact, so In fact is the best transition.

29 Choice A Transitions

The sentence describes how reception changed over time, so Over the years is the logical transition.

30 Choice D Transitions

The storm is an unfortunate event that contradicts the planning, so Unfortunately is the best transition.

31 Choice B Rhetorical synthesis

The unusual behaviour is the extraordinary memory for hidden seeds, which only the first choice states.

32 Choice C Rhetorical synthesis

Significance is expressed by the credited influence on conservation, which the first choice states along with the book's identity.

33 Choice D Rhetorical synthesis

The cause and effect are the rain, the rising river and the closed road, which the first choice links. The others pair unrelated facts.

Answer explanations

SAT Practice Test 5, Math, Module 1

Questions 1 to 14

1 **Choice C** **Data analysis**

Team 3 has 42 and Team 2 has 9, and 42 minus 9 is 33.

2 **Choice A** **Linear equations**

Subtract 3x from both sides: 3x + 7 = 22. Then 3x = 15, so x = 5.

3 **Choice D** **Linear equations in context**

The fee is paid once, so it is the constant 27. The $23 repeats, so it multiplies m. The total is 23m + 27.

4 **Choice B** **Systems of equations**

Adding the equations gives 2x = 8, so x = 4.

5 **Choice B** **Functions**

Each step of 1 in x adds 2 to f(x), so the slope is 2. Then f(1) = 8 gives 2 + 6, so the intercept is 6.

6 **Choice C** **Nonlinear expressions**

Multiply the coefficients, 3 times 4 is 12, and add the exponents, 5 plus 3 is 8.

7 **Choice A** **Ratios and rates**

27 cups of sugar is 9 times 3 cups, so the flour is 9 times 5 cups, which is 45.

8 **Choice D** **Triangles**

The angles of a triangle add to 180°, so 2x + 3x + 4x = 9x = 180 and x = 20.

9 **Choice C** **Circles**

The circumference $2\pi r = 22\pi$ gives r = 11. The area is $\pi r^2 = 121\pi$.

10 **Choice B** **Quadratics**

The equation factors as (x − 4)(x + 7) = 0, so x = 4 or x = −7.

11 **Choice D** **Probability**

39 students travel that way and 19 of them are in Grade 10, so the probability is 19/39. The fraction with 134 on the bottom uses every student.

12 **Choice A** **Statistics**

The first 6 scores total 492, and adding 96 gives 588. Divided by 7, the mean is 84.

13 **Choice C** **Scatterplots**

Substitute x = 8: 4(8) + 39 = 32 + 39 = 71.

14 **Choice D** **Inequalities**

The total is the 44 already joined plus 25 each week, and it must be at least 120, so 44 + 25w ≥ 120.

Answer explanations

SAT Practice Test 5, Math, Module 1

Questions 15 to 27

15 **Choice A** Functions

f(5) = 3(25) − 10 + 4 = 75 − 10 + 4 = 69.

16 **Choice B** Exponential functions

Tripling every 6 hours means the number of triplings is t/6, so the population is 800 times 3 to the power t/6.

17 **Choice C** Linear functions

The slope is (18 − 12)/(6 − 3) = 2. Using (3, 12), 12 = 2(3) + b gives b = 6.

18 **Choice D** Percent change

The increase is 900, and dividing by the original 6,000 gives 0.15, which is 15%. Dividing by the new value would give a smaller percent.

19 **Choice A** Volume

The volume is $\pi r^2 h = \pi(16)(12) = 192\pi$.

20 **Choice B** Right triangles

$7^2 + 24^2 = 625$, so the hypotenuse is $\sqrt{625} = 25$.

21 **10** Quadratics

The equation factors as (x − 10)(x + 2) = 0, so the solutions are 10 and −2. The positive one is 10.

22 **2** Linear equations

Expanding gives 5x − 25 = 6x − 27. Collecting terms gives −x = −2, so x = 2.

23 **48** Linear equations

Let the number be n. Then (2/3)n = 32, so n = 32 times 3/2 = 48.

24 **25** Systems of equations

Adding x + y = 30 and x − y = 20 gives 2x = 50, so the larger number is 25.

25 **15** Right triangles

By the Pythagorean theorem, the other leg is $\sqrt{(289 - 64)} = \sqrt{225} = 15$.

26 **15** Percent change

The increase is 12 dollars, and 12 divided by the original 80 is 0.15, which is 15 percent.

27 **396** Volume

The volume is $\pi r^2 h = \pi(36)(11) = 396\pi$, so k = 396.

Answer explanations

SAT Practice Test 5, Math, Module 2

Questions 1 to 14

1 **Choice D** Percentages

40% of 300 is 0.4 times 300, which is 120.

2 **Choice A** Volume

The volume is $\pi r^2 h = \pi(9)(6) = 54\pi$.

3 **Choice C** Inequalities

The total is the 54 already joined plus 15 each week, and it must be at least 200, so $54 + 15w \geq 200$.

4 **Choice B** Statistics

The first 5 scores total 310, and adding 80 gives 390. Divided by 6, the mean is 65.

5 **Choice A** Functions

$f(4) = 4(16) - 8 + 4 = 64 - 8 + 4 = 60$.

6 **Choice B** Probability

40 students travel that way and 19 of them are in Grade 10, so the probability is 19/40. The fraction with 123 on the bottom uses every student.

7 **Choice D** Systems of equations

Adding the equations gives $2x = 28$, so $x = 14$.

8 **Choice C** Equivalent expressions

The numerator factors as $(x - 6)(x - 5)$. Cancelling $x - 6$ leaves $x - 5$.

9 **Choice D** Ratios and rates

The pumps fill 1/12 and 1/4 of the tank each hour, together 1/3 of it, so the time is 3 hours.

10 **Choice C** Absolute value

The inequality means x is within 4 of 5, so $1 \leq x \leq 9$. That range has 9 integers, counting both ends.

11 **Choice A** Percentages

120 is 80 more than 40, and $80 / 40 = 2$, so $p = 200$.

12 **Choice B** Statistics

Multiplying by 2 multiplies the mean and the standard deviation by 2; adding 11 moves the mean by 11 but does not change the spread. The mean of A is 6, so the mean of B is 23, which is 17 greater. The standard deviation of B is 2 times that of A.

13 **Choice C** Systems of equations

A system has no solution when the lines are parallel but not identical. The second equation is 3 times the left side of the first with a different constant, so a must equal $6/3 = 2$.

14 **Choice A** Linear functions in context

The coefficient of x is the change in cost for each extra hour. The constant 30 is the fixed fee, and the 1 hour cost would be 34.2.

Answer explanations

SAT Practice Test 5, Math, Module 2

Questions 15 to 27

15 **Choice B** Surface area

Each cube has 6 faces, and gluing two faces together hides 2 of the 12, leaving 10 faces of area s^2. So $10s^2 = 640$, $s^2 = 64$, and $s = 8$.

16 **Choice D** Functions

g(2) = 7, so f(g(2)) = f(7) = 34. Also f(2) = 14, so g(f(2)) = g(14) = 199. The difference is 34 − 199 = −165.

17 **Choice D** Exponential equations

Write 4 as 2^2, so the right side is 2^(2x − 2). Equal bases give x + 4 = 2x − 2, so x = 6.

18 **Choice A** Percentages

Start at 100. After the 50% rise the price is 150, and 20% off 150 leaves 120. That is 20% higher than the original.

19 **Choice C** Exponential functions

A year is 12 months, so the value is multiplied by 0.9^(12/6) = 0.9^2 = 0.81 each year. That is a decrease of 1 − 0.81, or 19%.

20 **Choice B** Equivalent expressions

$(3x + 2)^2 = 9x^2 + 12x + 4$, and $(x − 4)^2 = x^2 − 8x + 16$. Subtracting the second from the first gives $8x^2 + 20x − 12$.

21 **28** Systems of equations

Adding x + y = 32 and x − y = 24 gives 2x = 56, so the larger number is 28.

22 **36** Quadratics

The maximum is at t = −b/(2a) = 8/2 = 4. Then h(4) = −1(16) + 32 + 20 = 36.

23 **4** Systems of equations

Adding the equations gives 7x + 7y = 28. Since the coefficients are equal, 7(x + y) = 28, so x + y = 4.

24 **60** Similar triangles

The scale factor is DE / AB = 12 / 3 = 4. The perimeter of ABC is 15, so the perimeter of DEF is 4 × 15 = 60.

25 **7** Circles in the plane

Complete the square: $(x + 2)^2 + (y + 6)^2 = 9 + 4 + 36 = 49$. The radius is √49 = 7.

26 **6** Quadratics

Exactly one real solution means the discriminant is zero: $b^2 − 4(9) = 0$, so $b^2 = 36$ and b = 6 (taking the positive value).

27 **5** Systems with nonlinear equations

Setting the expressions equal gives $x^2 − 4x + (9 − k) = 0$. One solution means the discriminant is 0: 16 − 4(9 − k) = 0, so k = 5.